

A singleton cyclic-sieving instance for Conjecture 426

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Theorem 1. *There is a cyclic-sieving triple with a prime-order rotation group and a polynomial that is equal to 1 at every root of unity of that order.*

Proof. Take two cyclically arranged positions and let $C = C_2$ act by their rotation. This is a group of order 2, which is prime. Let X consist of the single binary word 00, and put $X(q) = 1$. Rotating either zero or one step fixes this word. Thus, for every $g \in C$,

$$|X^g| = 1 = X(\omega(g)), \quad \omega(1_C) = 1, \quad \omega(\text{one-step rotation}) = -1.$$

The two values of ω are precisely the complex second roots of unity, and $X(q)$ has nonnegative integer coefficients. These are all the required cyclic-sieving equalities. Indeed, the polynomial is equal to 1 at every complex number. $\square$

The group rotates the two positions faithfully, while its action on X is trivial. The usual definition of a cyclic-sieving triple does not require a faithful action on X [1]. This distinction is essential here and is not concealed by the construction.

Scope of the statement. Both language versions of the conjecture ask for existence and impose no condition $|X| > 1$. In fact that additional condition would be impossible: 1 is one of the p -th roots of unity and the identity element fixes every member of X , so the two stipulated equalities give $|X| = X(1) = 1$. Thus the parenthetical description of a separation from “nontrivial counting” cannot mean a cardinality greater than one. We prove the explicit existence assertion as written and also explain this unavoidable limitation.

Formal verification. The Lean project defines the two-element cyclic group by Boolean exclusive-or and checks its full group laws, enumeration, generator, and prime order. Words are actual functions from the two positions to the binary alphabet. The action is actual permutation of positions; closure of X and both action laws are proved. A Boolean equality test is proved equivalent to function equality, so `fixedCount` counts the genuine fixed words of the complete, duplicate-free list X .

The polynomial is the coefficient list [1], evaluated by Horner’s rule. The theorem `polynomial_one_at_every_point` holds over every type supplied with the elementary evaluation operations and the laws $q0 = 0$, $q + 0 = q$, and the natural-number image of 1. In particular it holds over the complex numbers, with no restriction on the evaluation point. Hence this is not an evaluation at a finite sample of roots. The concrete root map into the integers checks that 1, -1 multiply according to C_2 and are distinct; the same values embed into the complex numbers. The universal evaluation theorem proves all cyclic-sieving equalities under that embedding. The file also proves that the identity equality forces cardinality one. No custom axioms, admitted proofs, or external computation oracles are used.

References

- [1] V. Reiner, D. Stanton and D. White, *The cyclic sieving phenomenon*, J. Combin. Theory Ser. A 108 (2004), 17–50, Definition–Proposition in Section 1. doi:10.1016/j.jcta.2004.04.009.